



Bricks

A LEGO® BRICK MODEL OF

Surprised Pikachu

The surprised face, with ears that swing, in 50 pieces.

50 pieces · 10 steps · 2 sections

Unofficial fan-made building instructions

Not physically build-tested. Every part is a real LEGO element and the model was checked on a computer (connections, collisions, every step, balance), but nobody has built it with real bricks yet.



Before you start

An unofficial fan-made manual

This booklet is not a LEGO product. It was generated from the model file by a computer program and checked part by part.

Another builder's design: its plans do not name them. If it is yours, tell us and we will credit you. These steps were worked out from it here.

Not physically build-tested

The model and these steps were checked on a computer only: every part/colour is a real LEGO element, all 84 connections between parts were matched in 3D, no two parts overlap, every part can be pushed into place in its step, and the model's balance was estimated. Not tested: how firmly parts hold, part tolerances and how it copes with handling. Take your time and support the model while you work.

The finished model

About 8.9×2.7 cm and 5.8 cm tall, about 25 g (estimated).

Almost nothing here is stacked: the face is built forward off a bracket standing on its edge, so its studs point at you. Start with the bracket, its three plates and the top of the head, then the eyes, nose and cheeks. Turn it round for the plates and bricks that hold the back together, and turn it back for the mouth. Last come the ears: two handles behind the top of the head, a clip plate on each, and the black tips.

You will need

- The 50 pieces in 4 colours listed at the back, with their LEGO element IDs and BrickLink numbers.
- A brick separator for pieces that sit too deep for a finger.

Tips

- Sort the pieces by colour before you start.
- Build the sections in order: each sub-assembly is finished before it is used.
- New pieces in a step have a yellow outline; earlier pieces are drawn paler.

The 4 colours

 Yellow: 34
 Black: 11

 Coral: 3
 Red: 2

How to read the steps

Parts and step number

7

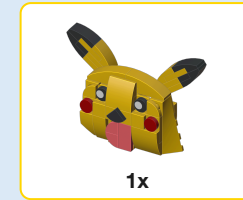


The parts for the step and how many of each, then the step number. The parts of a step are drawn at one scale, so a longer one looks longer; a tile or a long plate also has its size in studs under it.

New parts

The parts added in a step have a yellow outline and are drawn in full colour; the parts from earlier steps are drawn paler.

Sub-assemblies



A yellow-framed picture means: put in the sub-assembly you built in its own section. "Build 4x" means you need four of them.

Sections

Each section starts with its name at the top of the page. Sections are built in the order they appear.

Seen from below

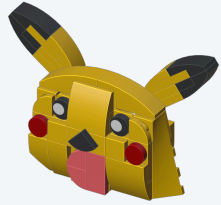
When a step adds parts underneath, the picture shows the model from below. A small label says so (and "seen from behind" likewise). A piece the picture cannot show gets a small picture beside it, ringed in yellow, from the side it can be seen from: where it clicks in.

Insert direction

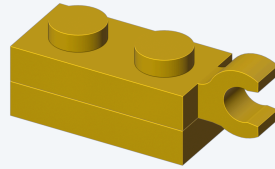
Some parts slide in sideways (axles through bearings, gears onto axles). The picture shows where they end up; push them in along their axle.

Build overview

The model is built in 2 sections. Each is finished before it is used.



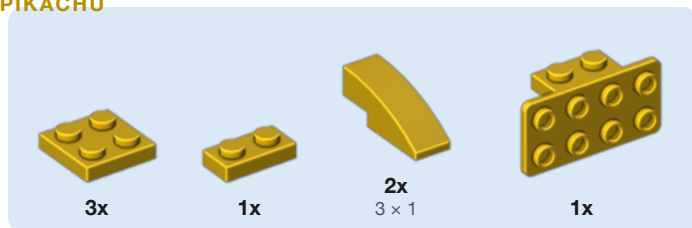
1. Surprised Pikachu
steps 1–7, 9–10



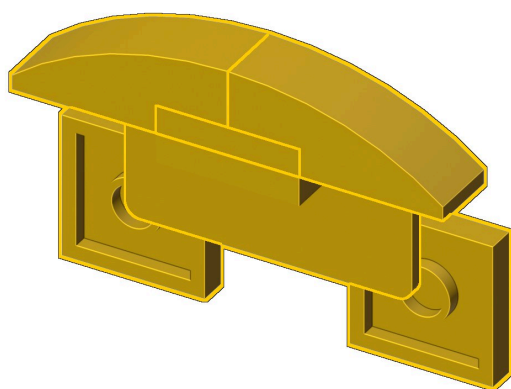
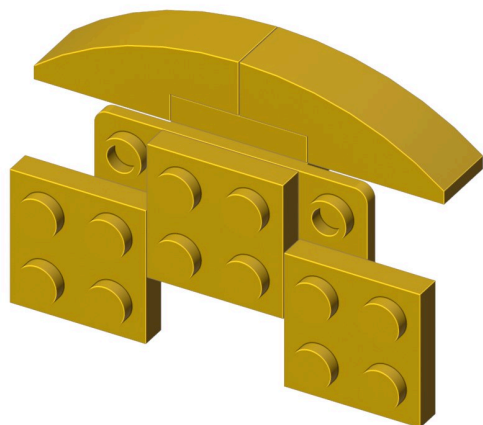
2. Ear base x2
step 8

SURPRISED PIKACHU

1

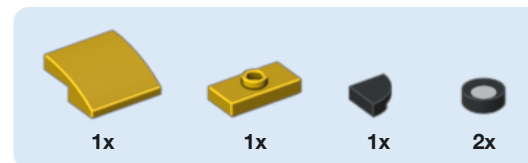


A bracket with three 2 x 2 plates on its upright side, and the top of the head

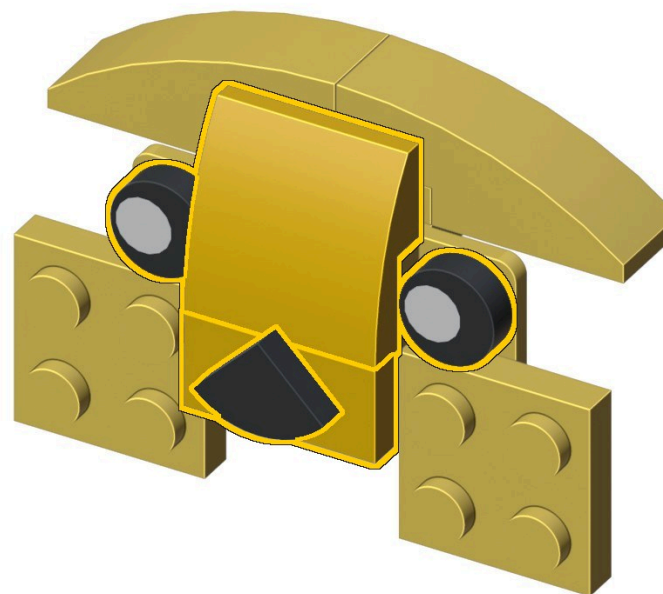


From behind: the outlined pieces.

2

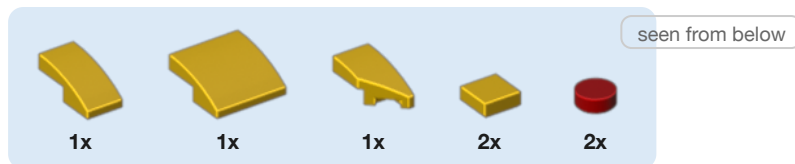


The eyes, the bridge of the nose and the nose

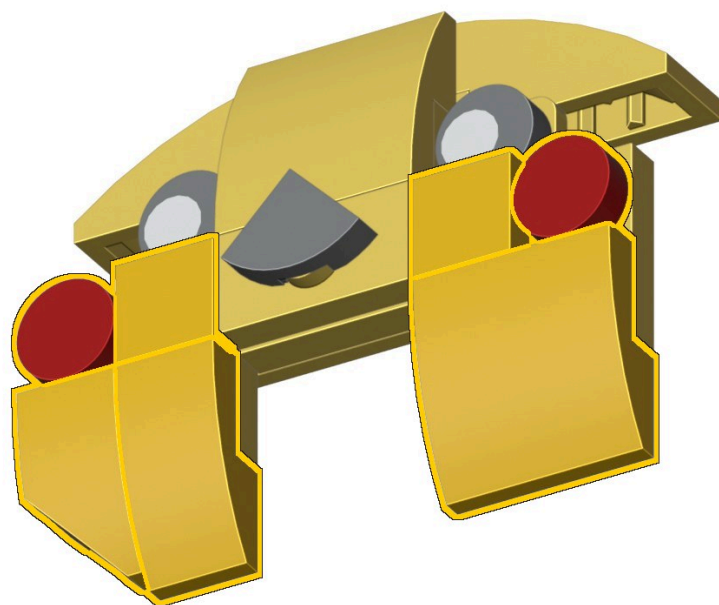


Surprised Pikachu

3

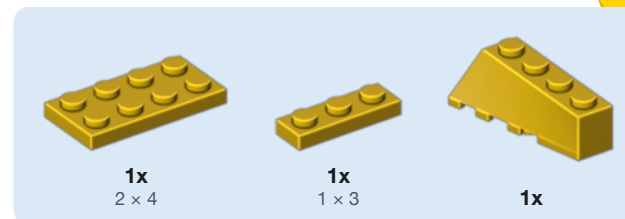


Red cheeks, and the curve under each one

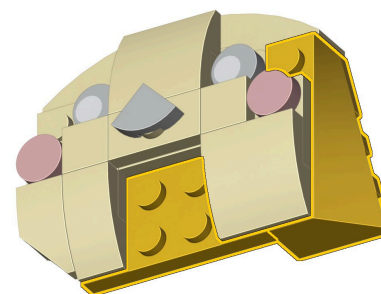
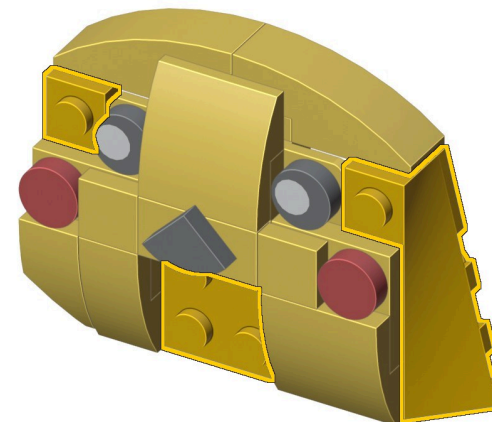


4

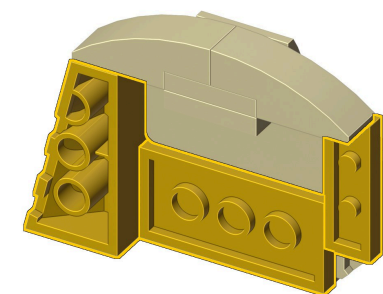
Surprised Pikachu



Turn it round: plates behind the face, and the sloping side of the head

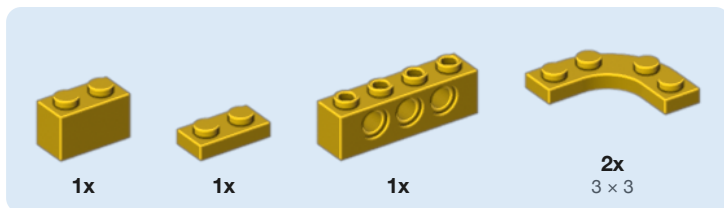


From below: the outlined pieces click in here.

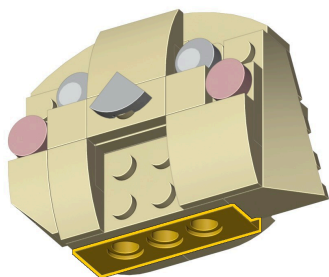
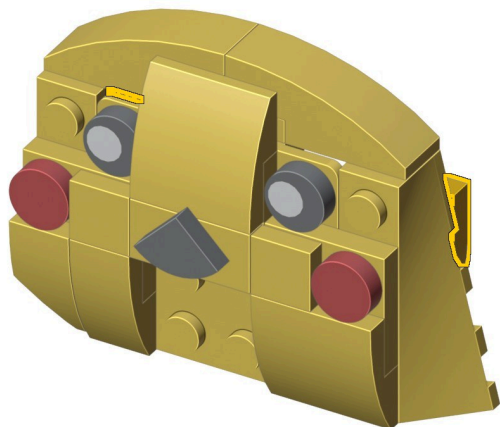


From behind: the outlined pieces.

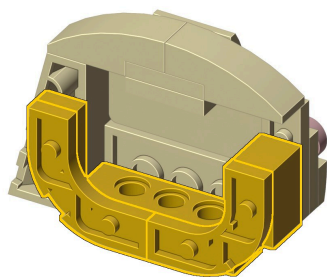
5



Bricks behind those, and two round corner plates across them all: the jaw



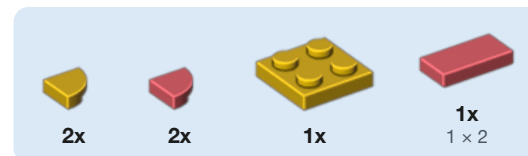
From below: the outlined piece clicks in here.



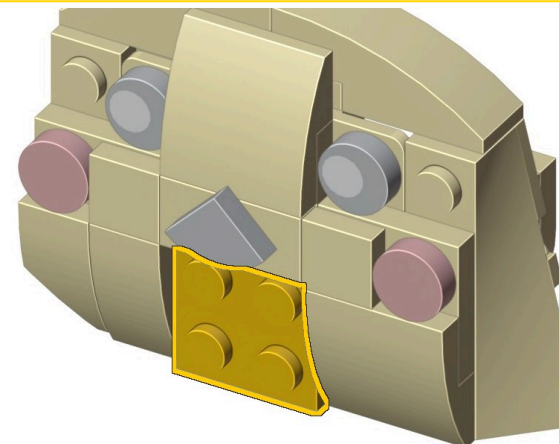
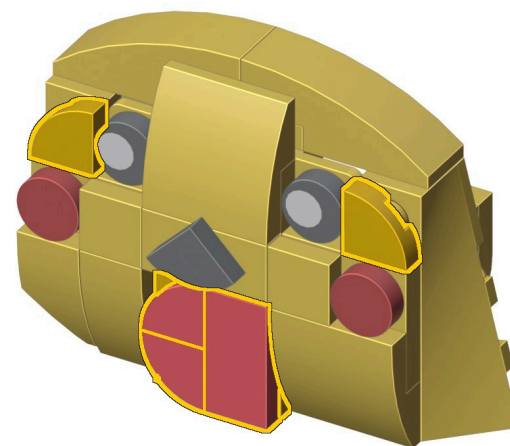
From behind: the outlined pieces.

6

Surprised Pikachu



Turn it back: a rounded corner beside each eye, and the open mouth

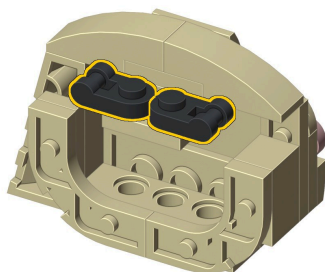
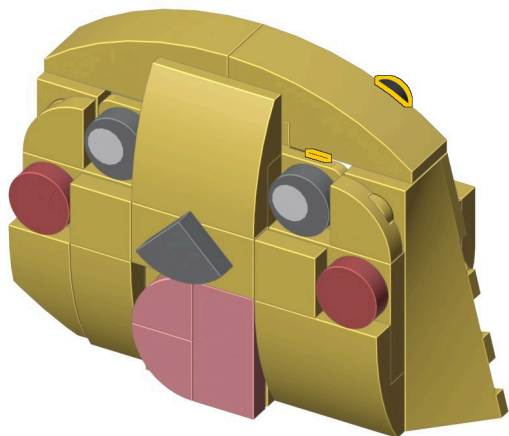


First the outlined piece: the others go on over it.

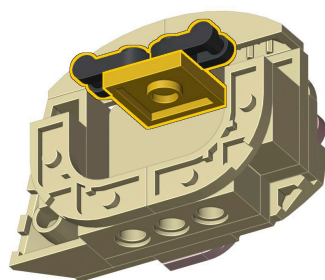
7



The ears' hinges: a plate pressed up under the top of the head, and two handles



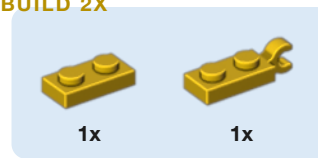
From behind: the outlined pieces.



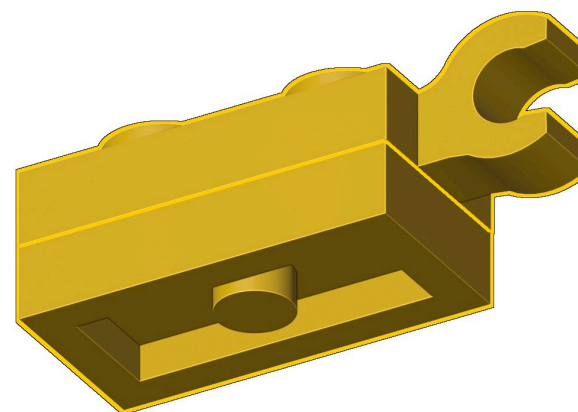
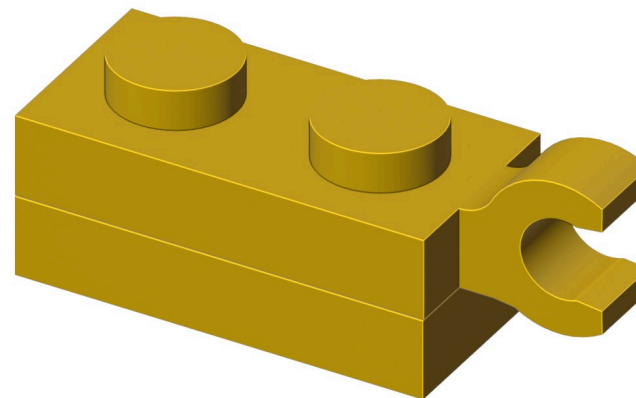
From behind and below: the outlined pieces.

EAR BASE · BUILD 2X

8



An ear's base: a clip plate with a plate under it. Make two

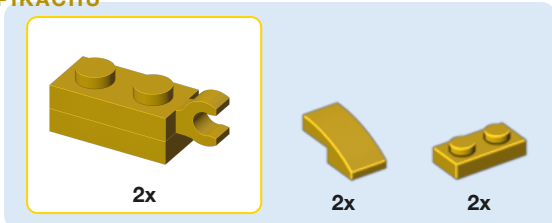


From below: the outlined pieces click in here.

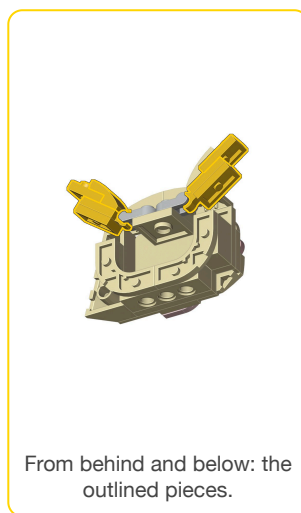
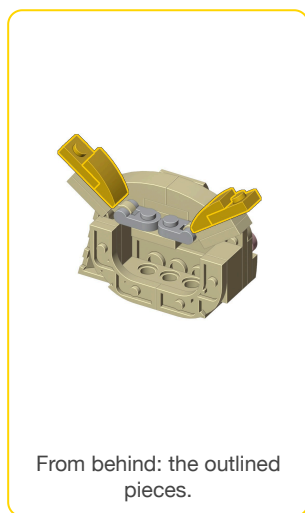
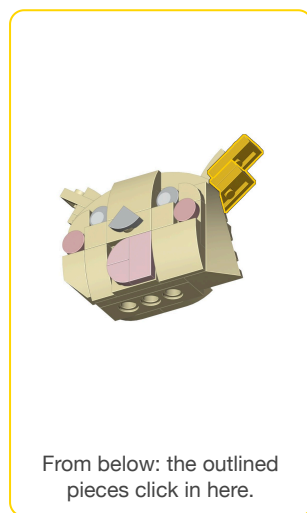
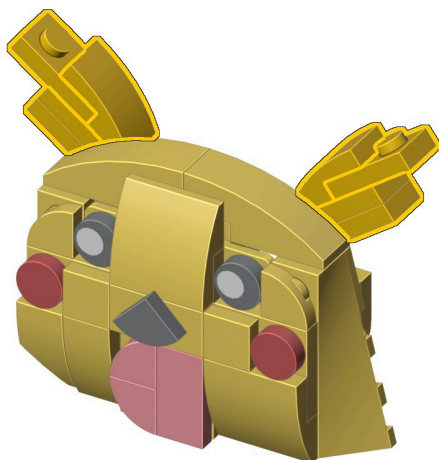
Ear base

SURPRISED PIKACHU

9



Clip an ear's base to each handle; add a plate over its end and a curved slope

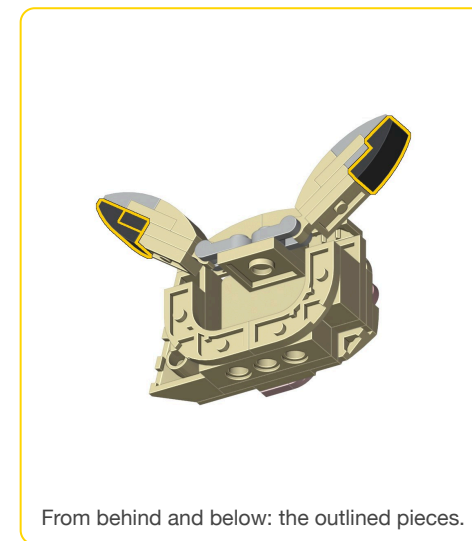
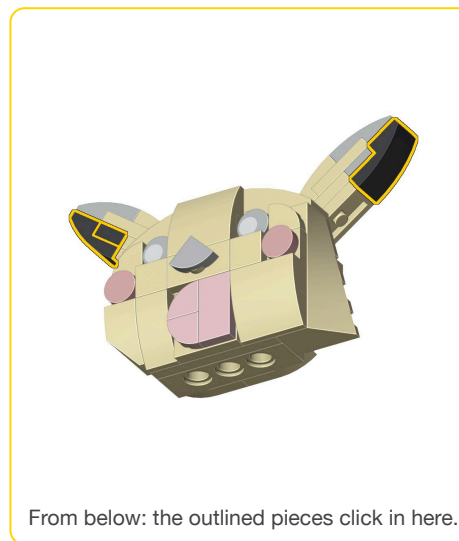
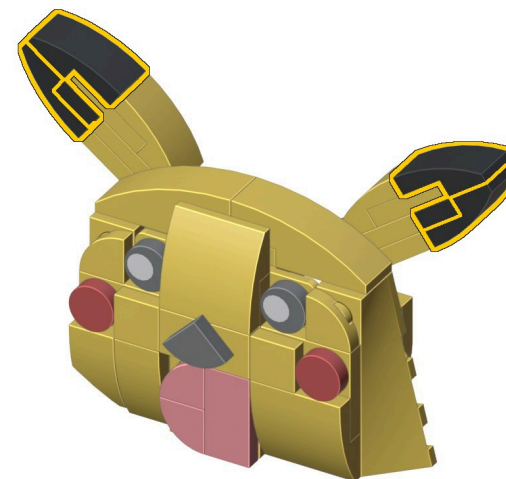


Surprised Pikachu

10

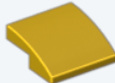
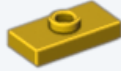
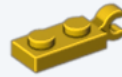
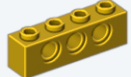


The black tips: a plate, a curved slope over it and an upside-down one under it



Parts inventory

50 pieces in 26 part/colour lines. Bold: the LEGO element ID (what LEGO Pick a Brick sells); below: the colour and the BrickLink part number.

 2x 6047276 Black BrickLink 11477	 2x 6147050 Black BrickLink 24201	 2x 302426 Black BrickLink 3024	 2x 6157554 Black BrickLink 26047	 1x 6172383 Black BrickLink 25269	 2x 6284079 Black BrickLink 98138pb074	 1x 6258566 Coral BrickLink 3069	 2x 6250022 Coral BrickLink 25269
 2x 6284574 Red BrickLink 98138	 1x 6118828 Yellow BrickLink 93274	 1x 4613966 Yellow BrickLink 3004	 3x 6029947 Yellow BrickLink 11477	 2x 6079007 Yellow BrickLink 15068	 2x 4247771 Yellow BrickLink 50950	 1x 6289743 Yellow BrickLink 29119	 1x 4262782 Yellow BrickLink 43720 <i>rare / discontinued</i>
 6x 302324 Yellow BrickLink 3023	 1x 362324 Yellow BrickLink 3623	 5x 4613978 Yellow BrickLink 3022	 1x 302024 Yellow BrickLink 3020	 2x 6308390 Yellow BrickLink 68568	 1x 6092583 Yellow BrickLink 15573	 2x 6347284 Yellow BrickLink 63868	 1x 370124 Yellow BrickLink 3701
 2x 307024 Yellow BrickLink 3070b	 2x 6195183 Yellow BrickLink 25269						

About this model

Almost nothing here is stacked: the face is built forward off a bracket standing on its edge, so its studs point at you. Start with the bracket, its three plates and the top of the head, then the eyes, nose and cheeks. Turn it round for the plates and bricks that hold the back together, and turn it back for the mouth. Last come the ears: two handles behind the top of the head, a clip plate on each, and the black tips.

Checked on a computer

WARN	real elements	26 part/colour combinations: 0 not real, 1 rare
PASS	connections	84 connections (82 stud, 2 clip); 1 separate piece(s)
PASS	collisions	0 overlapping pair(s) among 50 parts
PASS	buildability	46 insertions tried across 2 submodel(s); 0 problem(s)
PASS	stability	about 25 g; centre of mass inside the base (100% margin); tips at 25.3 deg
PASS	mechanism	no moving parts defined
PASS	electrics	no electrics in this model
PASS	technique	0 note(s)

Not yet built with real bricks

These checks catch most mistakes a designer can make, but not all: clutch power, part tolerances, how stiff a long axle is and how the model survives handling can only be judged with real bricks. If you build it, we would love to hear how it went.

Files

3D model (LDraw, opens in BrickLink Studio, LeoCAD and LDCad), parts lists for BrickLink and LEGO Pick a Brick, and an interactive 3D viewer:

bricks.superfun.games/m/pikachu

Trademarks

LEGO® is a trademark of the LEGO Group, which does not sponsor, authorize or endorse this booklet. Other names belong to their owners. Part geometry from the LDraw library (CC BY 4.0); connection data from the LDCad shadow library; element data from Rebrickable.